

A Inputs and freeze

Model-agnostic inputs

Observed response + model predictions + controls
perturbation labels + context labels

Provenance freeze

splits | genes | targets | preprocessing
checkpoint | code

B Audit modules

1 Global-fit audit

Raw-space agreement
Audit-delta agreement
Pearson | Spearman
RMSE | cosine

2 Perturbation-specific audit

Identify intended target
Rank among candidates
Top1 | Top5 | MRR

3 Falsification audit

Remove or scramble
target information
Mean-effect baseline
Blind/shuffled probes

4 Transfer and error-burden audit

Same targets across contexts
Within -> cross context
UER@K sensitivity
Sign-flip rate

C Claim profile

Endpoint-specific claim profile

Global fit
supported

Specificity
supported / narrowed

Transfer
supported / degraded

Error burden
sensitivity-only

From model performance to falsifiable claim boundaries